Solubility Curves

What a solubility curve actually reports, and why the same point on the graph can describe two different flasks.

Name ______________________ Date ______________________ Period ______________________

Demonstration: andresdbp.github.io/resources/demos/solubility-curves/

1 Before you open anything

Answer from what you already know. Being right is not the point — having something on paper to argue with later is.

PREDICTION

110 g of potassium nitrate, KNO_3, is stirred into 100 g of water at 60 °C and all of it dissolves. The flask is then cooled to 20 °C. How many grams of solid KNO_3 end up on the bottom?

Write one or two sentences saying what you assumed to get that number.

2 Four numbers, not one

Open the demonstration and press **Reset**. You should have KNO_3 selected, Temperature 60 °C, Solute added 110 g, Water 100 g, and **Stir and seed while cooling** ticked. The panel *State of the solution* reports four quantities and a coloured label.

1. With the Reset settings still in place, fill in the first row of the table.
2. Leave Temperature at 60 °C. Drag **Solute added** to **60 g** and fill in the second row.
3. Drag **Solute added** to **180 g** and fill in the third row.
4. Drag **Solute added** back to **110 g**, then drag **Temperature** down to **20 °C** and fill in the last row.

T (°C)	Solute added	Can dissolve	Dissolved	Room left	Undissolved solid	State (the coloured label)
60	110 g					
60	60 g					
60	180 g					
20	110 g					

a. In the first three rows you changed how much solute you put in, and one of the four readouts did not move at all. Which one, and why not?

b. Rows one and three have the same *Can dissolve* value but different states. Say in one sentence what the difference between the two flasks is.

c. Two of the four readouts always add up to a third. There are two such relationships here. Write both as equations, and check them against row three.

> The *Measured points* panel lists the handbook values the curve is drawn through. The line between those points is a monotone interpolation, so it never invents a bump the data do not show — but only the listed temperatures are measured. Everything between them is an estimate.

3 What temperature does to the ceiling

Keep Water at 100 g for all of this, so *Can dissolve* is read directly in grams per 100 g of water. Only the **Temperature** slider moves.

1. With KNO_3 selected, set Temperature to each value in the table and record **Can dissolve**.
2. Switch the solute to **NaCl**, then to **NH_3 (g)**, and fill the second table the same way. The NH_3 slider stops at 60 °C.

Temperature (°C)	KNO_3 — Can dissolve (g per 100 g H_2O)
0	
20	
40	
60	
80	
100	

Temperature (°C)	NaCl	NH_3 (g), at 1 atm NH_3
0		
20		
40		
60		

a. State in your own words what temperature does to the ceiling for each of the three solutes. One sentence each.

b. Between 0 and 60 °C, KNO_3 and NaCl both go up. Give the increase in g per 100 g of water for each. Why would nobody try to purify NaCl by dissolving it hot and cooling it?

c. Now settle Part 1. Using your KNO_3 table, calculate the mass of solid you expect on the bottom after cooling 110 g of KNO_3 in 100 g of water from 60 °C to 20 °C. Show the subtraction.

d. Press **Reset**, then press **Cool it → 20 °C**. Write down the mass the message reports, and compare it with your Part 1 prediction. If they differ, name the specific assumption that did not hold.

4 Working backwards, and one shortcut that fails

Exam questions rarely hand you the temperature. They hand you a mass and ask what conditions it needs — and they rarely use exactly 100 g of water.

A. Find the temperature.

1. Press **Reset**. Set **Solute added** to **150 g**, leaving Water at 100 g. Some of it is on the bottom.
2. Before touching the temperature: using your Part 3 table, estimate by interpolation the temperature at which 150 g would just dissolve. Show the working.
3. Now raise **Temperature** one degree at a time until **Undissolved solid** first reads 0.0 g. Record that temperature and the **Can dissolve** value there.

B. When the water is not 100 g.

1. Press **Reset**. Set **Water** to **150 g**, **Temperature** to **20 °C**, and **Solute added** to **60 g**.
2. Predict *Can dissolve* and *Undissolved solid* on paper first, then read them off.

	Your prediction (g)	What the demonstration reads (g)
Can dissolve		
Undissolved solid		

a. The curve is labelled "g per 100 g H_2O." Write the one line of arithmetic that converts a value read off the curve into a mass for this flask.

C. Two flasks, one dot.

1. Press **Reset**. Untick **Stir and seed while cooling**.
2. Press **Cool it → 20 °C** and fill in the second row of the table below. The first row is the flask you already made in Part 2, row four — the same 110 g of KNO_3 in the same 100 g of water at the same 20 °C.
3. Now press **Disturb it** and write down what the message says.

How you got to 20 °C	Can dissolve	Dissolved	Undissolved solid	State
Cooled with stirring on				
Cooled undisturbed				

b. What did **Disturb it** report?

c. Both flasks plot at the same point on the graph. Explain why the graph alone cannot tell you which of the two you are holding, and state what does determine it.

> A point plotted above the curve is not by itself evidence of supersaturation. Every flask in Part 2 row three and row four also plots above the curve, and none of them is supersaturated — they simply have solute that never dissolved.

5 Solve these without the demonstration

Close the page or look away from it. Show your working. Then reopen it and check yourself — and where you were wrong, write down where the reasoning went off. Handbook values you need are given with each problem.

1. 30 g of KNO_3 is stirred into 50 g of water at 60 °C, where the solubility is 110.0 g per 100 g H_2O. (i) Is the solution saturated or unsaturated, and by how many grams? (ii) The flask is cooled to 20 °C, where the solubility is 31.6 g per 100 g H_2O. What mass of solid appears, assuming it is stirred throughout?

2. A solution of KCl is saturated at 80 °C using 250 g of water. Solubility of KCl: 51.3 g per 100 g H_2O at 80 °C, 34.2 g per 100 g H_2O at 20 °C. What mass of KCl crystallises when the solution is cooled to 20 °C with stirring? (When you check this on the demonstration, the Solute added slider takes whole grams only, so set it to the nearest gram and expect agreement to within a few tenths.)

3. Ammonia dissolved in 100 g of water under 1 atm of NH_3 gas: 52.9 g at 20 °C, 23.5 g at 50 °C. (i) A saturated solution at 20 °C is warmed to 50 °C at constant NH_3 pressure. What mass of NH_3 leaves the solution? (ii) Explain why the NH_3 curve runs the opposite way to the KNO_3 curve. (iii) The curve is valid only at 1 atm of NH_3. Roughly what would the 20 °C value become at 0.5 atm, and what principle are you using?

4. Two sealed flasks each contain 110 g of KNO_3 and 100 g of water at 20 °C, so both plot at the same point on the solubility chart. One is a saturated solution over a bed of solid; the other is supersaturated and completely clear. Explain how each flask could have been prepared, describe one observation that would tell them apart, and state what the graph can and cannot tell you here. No arithmetic is required.

6 On your own

Nothing here is graded. The demonstration does more than this worksheet uses.

- Select NaCl and tick **Zoom the y-axis to this solute**. The curve that looked flat is not quite flat. Decide whether the change from 0 to 100 °C is big enough to be useful for anything.
- Tap or drag directly on the chart to mix fresh samples. Find the temperature at which KNO_3 and KCl are equally soluble, and say which one you would choose below that temperature.
- Drag **Water** down to 25 g and watch what happens to the range of the Solute added slider. Work out why the demonstration caps it where it does.
- With NH_3 selected, notice that the **Heat it** button changes its target. Read the note under the Temperature slider and explain what stops the curve at 60 °C — the chemistry, or the data?